

# EARL LOUIS

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## EDUCATION

**University of Connecticut**  
*Bachelor of Science in Robotics Engineering*

*Storrs, CT*  
Expected December 2026

## TECHNICAL SKILLS

**Automation & Control:** Control Panel Design, Sensor Network Integration, P&ID Development, Wireless Communication

**Software & Tools:** Python, C, C++, Ignition SCADA/HMI (Certified), AVR Microchip Studio

**Systems:** Process Automation, IoT, Power Distribution, Embedded Systems, I2C, SPI, UART

## PROFESSIONAL EXPERIENCE

### Barnes Aerospace

*Windsor, CT*

*Control Systems Intern*

June 2025 – Present

- Reviewed PLC I/O designs and developed P&ID diagrams to support control panel design and equipment integration
- Designed and implemented automated monitoring and control loops across facility operations, including chemical process control, safety equipment tracking, spill detection, and environmental compliance systems
- Earned Ignition 8.1 Credential; configured HMI/SCADA displays providing operators real-time data visualization

### Norwich Public Utilities

*Norwich, CT*

*Electrical Engineering Intern*

June 2024 – January 2025

- Assisted in design and implementation of power distribution systems, ensuring compliance with industry standards
- Performed transformer testing and prepared phasing diagrams for high-voltage equipment installations
- Analyzed transformer load data to identify critical infrastructure needs and updated circuit diagrams accordingly
- Conducted field inspections supporting maintenance planning and system upgrades

### Outlier AI

*Remote*

*Machine Learning Quality Assurance Tester*

May 2024 – January 2025

- Evaluated Meta AI-generated responses across technical, conversational, and health-related content areas, grading accuracy, relevance, and adherence to safety guidelines
- Flagged and downgraded responses where the model overstepped into medical or mental health advice, contributing to improved model safety

### Demers Exposition Services, Inc.

*Uncasville, CT*

*Stage Technician*

December 2022 – June 2024

- Coordinated setup and teardown of large-scale arena events, providing technical support for audio, lighting, video, and rigging systems

## FEATURED PROJECTS

### Drone Health Monitoring System

*Raspberry Pi 4, I2C/SPI/UART, FFT, 915 MHz Radio*

- Joint senior design project. Fitted a quadcopter with current, vibration, and temperature sensors. An onboard Raspberry Pi performed FFT-based processing and transmitted data over 915 MHz radio to a ground station running a live dashboard with automated out-of-range alarms.

### Two-Wheeled Path Planning Robot

*Romi 32U4, Raspberry Pi 3, Python, C++*

- Navigation robot with a custom wavefront path planner and overhead camera pipeline for real-time localization and orientation detection.

### Sound Intensity Visualizer

*AVR 128DB48, C (AVR Microchip Studio), LED Dot Matrix, OLED Display*

- Audio visualizer displaying live sound waveforms on an LED dot matrix with dB readout on a secondary OLED screen.